Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Need for statistics: purposes, and the collection and tabulation of data. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Third Term
Week	9
Topic	Need for statistic
Subtopic	Need for statistics: purposes, and the collection and tabulation of data
Theme	—
Curriculum objectives	• explain the purpose of statistic • analyse and interpret statistical data presented in tables
Previous knowledge	• Students are familiar with counting and recording information.
Lesson sequence	• explain the purpose of statistic — the teacher introduces the concept of statistics as a method of collecting and organising information • analyse and interpret statistical data presented in tables — the teacher takes the students through collecting data, recording it with tally marks, and setting it out in a frequency table
Instructional materials	• classroom register, bottle tops, pencils or counters.
Evaluation	• (a) Define statistic • (b) State three methods of collecting data. • (c) Prepare a frequency table for the following data: 1, 1, 3, 4, 4, 3, 2, 2, 3, 3, 4, 1, 2.

Lesson objectives

1.	explain the purpose of statistic
2.	analyse and interpret statistical data presented in tables

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Need for statistics: purposes, and the collection and tabulation of data

Class: JSS1 **Subject:** Mathematics **Week:** 9

Section 1: What Statistics Is For

Whenever you need to know something about a group — how many pupils are in your class, how many bottle tops you have, which scores your classmates got in a test — you are working with data. Data are facts or information collected for a particular purpose. The collection on its own is not enough: the information is only useful once you can see what it means.

Statistics is the science of collecting, organizing, presenting, analyzing and interpreting data. It is the whole journey your information takes, from the moment you gather it to the moment you understand what it tells you.

Statistics matters because it helps you to:

- Collect information — gather the facts you need.
- Organize information — arrange it so it is easy to look through.
- Compare information — see how one thing stands against another.
- Make decisions — choose sensibly from what the information shows.
- Present information clearly — show it so others can understand it too.

A single fact is just a fact. But when you collect facts, organize them and present them clearly, they become something you can compare and decide on. That is the purpose of statistics.

Practice

1. Define statistics in your own words.
2. What are data?
3. State two ways that statistics helps you.

Section 2: From Raw Data to a Frequency Table

Very often we want to answer a question about a group of things, like "which vehicle passes our school most often?" or "what is the favourite colour in our class?". To answer that, we first need data — that is, pieces of information we have collected. Data can be collected in a few different ways:

Four common ways to collect data are:

Observation — watching and noting what you see, for example watching the road to record passing vehicles.
Counting — counting the items you want to know about, for example counting the pencils on each desk.
Interview — asking people questions and recording their answers.
Questionnaire — giving people a written list of questions to answer.

Once you have collected data, you need a neat way to record it as it comes in. That is what tally marks are for. A tally is a small stroke | written for each item you record. When you reach the fifth stroke, you draw it across the other four, like this: ||||/ (which counts as 5).

When the tallying is finished, we can count the strokes for each item and write down how many times it occurred. A frequency table is a table that shows how many times each item occurs in a set of data. The number of times an item occurs is called its frequency. For example, the favourite colours of a group of students can be shown in a frequency table:

Colour	Frequency
Red	18
Blue	20
Green	14
Yellow	16
White	15
Pink	30

Let's put this all together with the data collected by watching vehicles pass:
car, car, car, car, car, car, bus, lorry, lorry, bicycle, bicycle, bicycle
We record each vehicle with a tally mark as it appears. The tally for "car" builds up as six strokes, which we group as five and one extra: |||| |, so its total is 6. The other vehicles follow the same way.

Now we read the totals from the tally to make a frequency table. The vehicle that occurs most often has the largest frequency, and the one that occurs least often has the smallest:

Vehicles	Tally	Frequency
car	|||||		6
bus			1
lorry	||	2
bicycle	|||	3

Reading the table tells us: car appeared 6 times, bus 1 time, lorry 2 times and bicycle 3 times. The car is the most common vehicle, and the bus is the least common.

Practice

1. List four different ways of collecting data.
2. Explain what tally marks are used for, and write the tally for 7 items.
3. Prepare a frequency table for the following data:
   1, 1, 3, 4, 4, 3, 2, 2, 3, 3, 4, 1, 2

Check your work

Attempt every question before you read this page.

What Statistics Is For

1. Statistics is the process of gathering facts, arranging them in an orderly way, showing them clearly, and using them to understand and make decisions.
2. Data are facts or information collected for a particular purpose.
3. Any two of: it helps you to collect information, organize information, compare information, make decisions, or present information clearly. (For example, it helps you to compare information and to make decisions.)

From Raw Data to a Frequency Table

1. Observation, counting, interview and questionnaire.
2. Tally marks are used to record data as it is collected. The tally for 7 is |||| || (one group of five and two extra strokes).

   Each item is one stroke |. When you reach five strokes you draw the fifth across the other four as |||| (which is 5). For 7 items, you make one group of five (||||) and then two more single strokes (||), giving |||| ||.

3.

Number	Frequency
1	3
2	3
3	4
4	3

Count how many times each number occurs. The number 1 appears in positions 1, 2 and 12 — that is 3 times. The number 2 appears in positions 7, 8 and 13 — that is 3 times. The number 3 appears in positions 3, 6, 9 and 10 — that is 4 times. The number 4 appears in positions 4, 5 and 11 — that is 3 times. Write each number once in the first column and its count in the frequency column.

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-40f8889b	**Date**	
Subtopic	Need for statistics: purposes, and the collection and tabulation of data	**Rater**	

> The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____